Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java

Practical XI

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Class:TYIT	Batch:1
Date of Assignment: 12-9-2026	Date/Time of Submission: 12-9-2026

Aim:-Develop a five page web application site using any two or three Java EE Technologies.

Code:-

Index.jsp:-

```
\<!DOCTYPE
html>
\<html>
\<head>
   \<title>Online Book
Store\</title>
   \<link rel="stylesheet"
href="style.css">
\</head>
\<body>
\<div
class="navbar">
   \<h2>Online Book Store\</h2>
   \<div>
      \<a
href="index.jsp">Home\</a>
      \<a
href="books.jsp">Books\</a>
      \<a
href="register.jsp">Register\</a>
      \<a
href="login.jsp">Login\</a>
      \<a
href="cart.jsp">Cart\</a>
   \</div>
\</div>
\<div
class="hero">
   \<h1>Welcome to Online Book
Store\</h1>
   \<p>
      Find your favourite books at affordable
prices.
   \</p>
   \<a href="books.jsp"
class="button">
      View Books
   \</a>
\</div>
\<div
class="container">
   \<h2>About Our Store\</h2>
   \<p>
      Online Book Store is a simple Java EE
web application
      where users can browse books, register,
login and add
      books to their shopping cart.
   \</p>
\</div>
\</body>
\</html>
**Books.jsp:-**
<%@
page import="java.sql.\*" %>
<%@
page import="com.bookstore.DBConnection" %>
\<!DOCTYPE
html>
\<html>
\<head>
   \<title>Books\</title>
   \<link rel="stylesheet"
href="style.css">
\</head>
\<body>
\<div
class="navbar">
   \<h2>Online Book Store\</h2>
   \<div>
      \<a
href="index.jsp">Home\</a>
      \<a
href="books.jsp">Books\</a>
      \<a
href="register.jsp">Register\</a>
```

```
        \<a
href="login.jsp">Login\</a>
        \<a
href="cart.jsp">Cart\</a>
    \</div>
\</div>
\<div
class="container">
    \<h1>Available Books\</h1>
    \<table>
        \<tr>
            \<th>ID\</th>
            \<th>Book Title\</th>
            \<th>Author\</th>
            \<th>Price\</th>
            \<th>Action\</th>
        \</tr>
        <%
            Connection con = null;
            PreparedStatement ps = null;
            ResultSet rs = null;
            try {
                con =
DBConnection.getConnection();
                ps =
con.prepareStatement("SELECT \* FROM books");
                rs = ps.executeQuery();
                while(rs.next()) {
        %>
        \<tr>
            \<td><%=
rs.getInt("id") %>\</td>
            \<td><%=
rs.getString("title") %>\</td>
            \<td><%=
rs.getString("author") %>\</td>
            \<td>₹ <%=
rs.getDouble("price") %>\</td>
            \<td>
                \<a class="button"
href="CartServlet?book_id=<%= rs.getInt("id")
%>">
                Add to Cart
                \</a>
            \</td>
        \</tr>
        <%
                }
            } catch(Exception e) {
                out.println("Error: "
\+ e.getMessage());
            } finally {
                try {
                    if(rs != null) rs.close();
                    if(ps != null) ps.close();
                    if(con != null)
con.close();
                } catch(Exception e) {
                }
            }
        %>
    \</table>
\</div>
\</body>
\</html>
**Cart.jsp:-**
<%@
page import="java.sql.\*" %>
<%@
page import="com.bookstore.DBConnection" %>
\<!DOCTYPE
html>
\<html>
\<head>
    \<title>Shopping Cart\</title>
    \<link rel="stylesheet"
href="style.css">
\</head>
\<body>
\<div
class="navbar">
    \<h2>Online Book Store\</h2>
    \<div>
        \<a
href="index.jsp">Home\</a>
        \<a
href="books.jsp">Books\</a>
        \<a
href="register.jsp">Register\</a>
        \<a
href="login.jsp">Login\</a>
        \<a
href="cart.jsp">Cart\</a>
    \</div>
\</div>
\<div
class="container">
    \<h1>Shopping Cart\</h1>
    <%
        Integer userId = (Integer)
session.getAttribute("user_id");
        if(userId == null) {
    %>
        \<p>
            Please login to view your cart.
        \</p>
        \<a href="login.jsp"
class="button">
```

```
            Login
        \</a>
    <%
        } else {
            Connection con = null;
            PreparedStatement ps = null;
            ResultSet rs = null;
            try {
                con =
DBConnection.getConnection();
                String sql =
                "SELECT books.title,
books.author, books.price, cart.quantity " +
                "FROM cart INNER JOIN
books ON cart.book_id = books.id " +
                "WHERE
cart.user_id=?";
                ps = con.prepareStatement(sql);
                ps.setInt(1, userId);
                rs = ps.executeQuery();
    %>
    \<table>
        \<tr>
            \<th>Book\</th>
            \<th>Author\</th>
            \<th>Price\</th>
            \<th>Quantity\</th>
        \</tr>
        <%
            boolean found = false;
            while(rs.next()) {
                found = true;
        %>
        \<tr>
            \<td>
                <%=
rs.getString("title") %>
            \</td>
            \<td>
                <%=
rs.getString("author") %>
            \</td>
            \<td>
                ₹ <%=
rs.getDouble("price") %>
            \</td>
            \<td>
                <%=
rs.getInt("quantity") %>
            \</td>
        \</tr>
        <%
            }
            if(!found) {
        %>
        \<tr>
            \<td colspan="4">
                Your cart is empty.
            \</td>
        \</tr>
        <%
            }
        %>
    \</table>
    <%
            } catch(Exception e) {
                out.println("Error: "
\+ e.getMessage());
            } finally {
                try {
                    if(rs != null) rs.close();
                    if(ps != null) ps.close();
                    if(con != null)
con.close();
                } catch(Exception e) {
                }
            }
        }
    %>
\</div>
\</body>
\</html>
**Contact.jsp:-**
\<!DOCTYPE
html>
\<html>
\<head>
    \<title>Contact -
BookHub\</title>
    \<link rel="stylesheet"
href="css/style.css">
\</head>
\<body>
\<div
class="header">
    \<div class="logo">
        BookHub
    \</div>
    \<div class="nav">
        \<a
href="index.jsp">Home\</a>
        \<a
href="books.jsp">Books\</a>
        \<a
href="login.jsp">Login\</a>
        \<a
href="cart.jsp">Cart\</a>
        \<a
```

```
href="contact.jsp">Contact\</a>
    \</div>
\</div>
\<div
class="container">
    \<div class="hero">
        \<h1>
            Contact Us
        \</h1>
        \<p>
            Have a question?
            We would love to hear from you.
        \</p>
        \<h3>
            Email
        \</h3>
        \<p>
            bookhub\@gmail.com
        \</p>
        \<h3>
            Phone
        \</h3>
        \<p>
            9876543210
        \</p>
        \<h3>
            Address
        \</h3>
        \<p>
            Mumbai, Maharashtra, India
        \</p>
    \</div>
\</div>
\<div
class="footer">
    \<p>
        © 2026 BookHub Online Book Store
    \</p>
\</div>
\</body>
\</html>
**Login.jsp:-**
\<!DOCTYPE
html>
\<html>
\<head>
    \<title>Login\</title>
    \<link rel="stylesheet"
href="style.css">
\</head>
\<body>
\<div
class="navbar">
    \<h2>Online Book Store\</h2>
    \<div>
        \<a
href="index.jsp">Home\</a>
        \<a
href="books.jsp">Books\</a>
        \<a
href="register.jsp">Register\</a>
        \<a
href="login.jsp">Login\</a>
        \<a
href="cart.jsp">Cart\</a>
    \</div>
\</div>
\<div
class="form-container">
    \<h1>Login\</h1>
    \<form action="LoginServlet"
method="post">
        \<label>Email\</label>
        \<input type="email"
               name="email"
               placeholder="Enter
email"
               required>
        \<label>Password\</label>
        \<input type="password"
               name="password"
               placeholder="Enter
password"
               required>
        \<input type="submit"
               value="Login"
               class="button">
    \</form>
    \<p>
        Don't have an account?
        \<a
href="register.jsp">Register\</a>
    \</p>
\</div>
\</body>
\</html>
**Register.jsp:-**
\<!DOCTYPE
html>
\<html>
\<head>
    \<title>Register\</title>
    \<link rel="stylesheet"
href="style.css">
\</head>
\<body>
\<div
class="navbar">
```

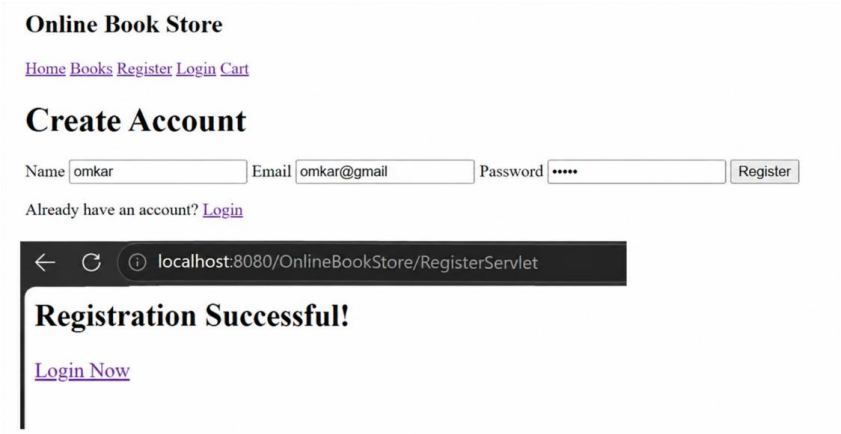
```
   \<h2>Online Book Store\</h2>
   \<div>
      \<a
href="index.jsp">Home\</a>
      \<a
href="books.jsp">Books\</a>
      \<a
href="register.jsp">Register\</a>
      \<a
href="login.jsp">Login\</a>
      \<a
href="cart.jsp">Cart\</a>
   \</div>
\</div>
\<div
class="form-container">
   \<h1>Create Account\</h1>
   \<form action="RegisterServlet"
method="post">
      \<label>Name\</label>
      \<input type="text"
            name="name"
            placeholder="Enter your
name"
            required>
      \<label>Email\</label>
      \<input type="email"
            name="email"
            placeholder="Enter your
email"
            required>
      \<label>Password\</label>
      \<input type="password"
            name="password"
            placeholder="Enter
password"
            required>
      \<input type="submit"
            value="Register"
            class="button">
   \</form>
   \<p>
      Already have an account?
      \<a
href="login.jsp">Login\</a>
   \</p>
\</div>
\</body>
\</html>
**CartServlet.java:-**
package
com.bookstore;
import
java.io.IOException;
import
java.sql.Connection;
import
java.sql.PreparedStatement;
import
javax.servlet.ServletException;
import
javax.servlet.annotation.WebServlet;
import
javax.servlet.http.HttpServlet;
import
javax.servlet.http.HttpServletRequest;
import
javax.servlet.http.HttpServletResponse;
import
javax.servlet.http.HttpSession;
@WebServlet("/CartServlet")
public
class CartServlet extends HttpServlet {
   protected void doGet(HttpServletRequest
request,
         HttpServletResponse response)
         throws ServletException,
IOException {
      HttpSession session =
request.getSession();
      Integer userId =
            (Integer)
session.getAttribute("user_id");
      if(userId == null) {
response.sendRedirect("login.jsp");
         return;
      }
      int bookId =
            Integer.parseInt(
request.getParameter("book_id")
            );
      try {
         Connection con =
DBConnection.getConnection();
         String sql =
               "INSERT INTO
cart(user_id,book_id,quantity) " +
               "VALUES(?,?,1)";
         PreparedStatement ps =
               con.prepareStatement(sql);
         ps.setInt(1, userId);
         ps.setInt(2, bookId);
         ps.executeUpdate();
         ps.close();
         con.close();
response.sendRedirect("cart.jsp");
      } catch(Exception e) {
         response.getWriter().println(
```

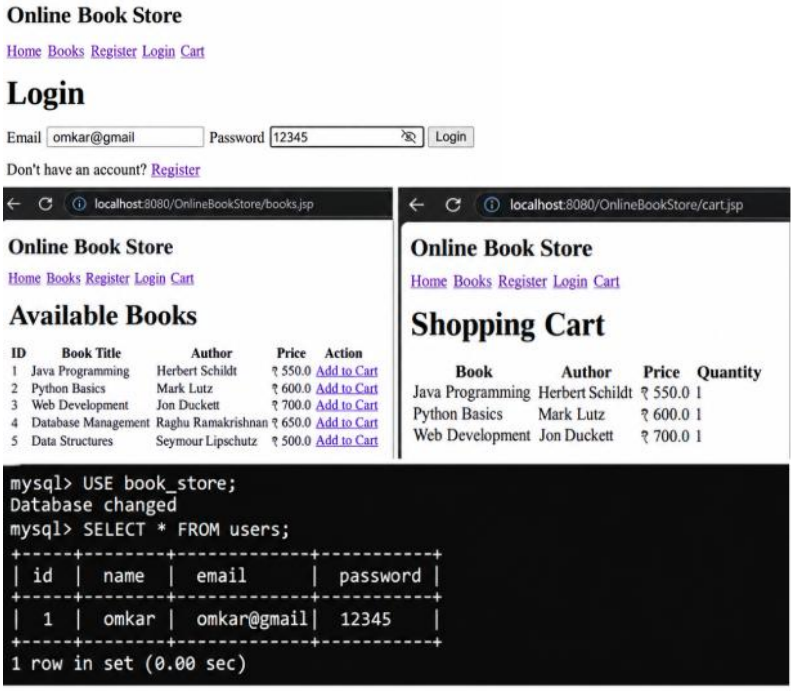
```
                "Error: " +
e.getMessage()
            );
        }
    }
}
**DBconnection.java:-**
package
com.bookstore;
import
java.sql.Connection;
import
java.sql.DriverManager;
public
class DBConnection {
    public static Connection getConnection()
throws Exception {
Class.forName("com.mysql.jdbc.Driver");
        Connection con =
DriverManager.getConnection(
"jdbc\:mysql://localhost:3306/book_store",
            "root",
            "root"
        );
        return con;
    }
}
**LoginServlet.java:-**
package
com.bookstore;
import
java.io.IOException;
import
java.sql.Connection;
import
java.sql.PreparedStatement;
import
java.sql.ResultSet;
import
javax.servlet.ServletException;
import
javax.servlet.annotation.WebServlet;
import
javax.servlet.http.HttpServlet;
import
javax.servlet.http.HttpServletRequest;
import
javax.servlet.http.HttpServletResponse;
import
javax.servlet.http.HttpSession;
@WebServlet("/LoginServlet")
public
class LoginServlet extends HttpServlet {
    protected void doPost(HttpServletRequest
request,
        HttpServletResponse response)
        throws ServletException,
IOException {
        String email =
request.getParameter("email");
        String password =
request.getParameter("password");
        try {
            Connection con =
DBConnection.getConnection();
            String sql = "SELECT \* FROM
users WHERE email=? AND password=?";
            PreparedStatement ps =
con.prepareStatement(sql);
            ps.setString(1, email);
            ps.setString(2, password);
            ResultSet rs = ps.executeQuery();
            if (rs.next()) {
                HttpSession session =
request.getSession();
session.setAttribute("user_id", rs.getInt("id"));
session.setAttribute("user_name",
rs.getString("name"));
response.sendRedirect("books.jsp");
            } else {
response.setContentType("text/html");
response.getWriter().println("\<h2>Invalid Email or
Password\</h2>");
response.getWriter().println("\<a href='login.jsp'>Try
Again\</a>");
            }
            rs.close();
            ps.close();
            con.close();
        } catch (Exception e) {
response.setContentType("text/html");
response.getWriter().println("\<h2>Error: " +
e.getMessage() + "\</h2>");
        }
    }
}
**RegisterServlet.java:-**
package
com.bookstore;
import
java.io.IOException;
import
java.io.PrintWriter;
import
java.sql.Connection;
import
java.sql.PreparedStatement;
import
```

```
javax.servlet.ServletException;
import
javax.servlet.annotation.WebServlet;
import
javax.servlet.http.HttpServlet;
import
javax.servlet.http.HttpServletRequest;
import
javax.servlet.http.HttpServletResponse;
@WebServlet("/RegisterServlet")
public
class RegisterServlet extends HttpServlet {
   protected void doPost(HttpServletRequest
request,
         HttpServletResponse response)
         throws ServletException,
IOException {
response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String name =
request.getParameter("name");
      String email =
request.getParameter("email");
      String password =
request.getParameter("password");
      try {
         Connection con =
DBConnection.getConnection();
         String sql =
               "INSERT INTO
users(name,email,password) VALUES(?,?,?)";
         PreparedStatement ps =
               con.prepareStatement(sql);
         ps.setString(1, name);
         ps.setString(2, email);
         ps.setString(3, password);
         int result = ps.executeUpdate();
         if(result > 0) {
out.println("\<h2>Registration Successful!\</h2>");
            out.println("\<a
href='login.jsp'>Login Now\</a>");
         } else {
out.println("\<h2>Registration Failed\</h2>");
         }
         ps.close();
         con.close();
      } catch(Exception e) {
         out.println("\<h2>Error:
" + e.getMessage() + "\</h2>");
      }
   }
}
```

Output:

```
Output:-
**SQL
CODE:-**
CREATE
DATABASE book_store;
USE
book_store;
CREATE
TABLE users (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(100) NOT NULL,
   email VARCHAR(100) UNIQUE NOT NULL,
   password VARCHAR(100) NOT NULL
);
CREATE
TABLE books (
   id INT PRIMARY KEY AUTO_INCREMENT,
   title VARCHAR(150) NOT NULL,
   author VARCHAR(100) NOT NULL,
   price DOUBLE NOT NULL
);
CREATE
TABLE cart (
   id INT PRIMARY KEY AUTO_INCREMENT,
   user_id INT NOT NULL,
   book_id INT NOT NULL,
   quantity INT DEFAULT 1
);
INSERT
INTO books (title, author, price) VALUES
('Java
Programming', 'Herbert Schildt', 550),
('Python
Basics', 'Mark Lutz', 600),
('Web
Development', 'Jon Duckett', 700),
('Database
Management', 'Raghu Ramakrishnan', 650),
('Data
Structures', 'Seymour Lipschutz', 500);
```






```
mysql> USE book_store;
Database changed
mysql> SELECT * FROM users;
+-----+--------+-------------+----------+
| id  | name   | email       | password |
+-----+--------+-------------+----------+
|  1  | omkar  | omkar@gmail |  12345   |
+-----+--------+-------------+----------+
1 row in set (0.00 sec)
```